

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XVI: THE OUT-COVERING CORRESPONDENCE OF THE HEAD MAP, ITS K_0 -INDEX η_* , AND THE KK -CATEGORY OF THE IWAHORI TOWER

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ABSTRACT. [VIII] realized the tail norm of the Iwahori tower on the shadow Kirchberg algebras $\mathcal{O}_k$: the tail map τ is an in-covering, so it induces a canonical unital $*$ -homomorphism $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, whose K_0 -map is the pullback τ^* up the tower. It left open [VIII, Rem. 2.5, 2.6, 5.1(a)] the mirror question: the head map η is an out-covering, the pushforward η_* is the one that descends on the K_0 -side ($\eta_* A_{k+1}^t = A_k^t \eta_*$), yet no $*$ -homomorphism $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ can induce it, because $\eta_*[1] = q[1]$ forces non-unitality and the Kirchberg–Phillips lift is non-canonical. This paper settles it: the canonical object is not a homomorphism but a C^* -correspondence. We build from the out-covering $\eta^\sharp$ a right Hilbert $\mathcal{O}_k$ -module $\mathcal{X}_\eta = \bigoplus_w p_{\eta^\sharp(w)} \mathcal{O}_k$, finitely generated and projective, and a left action $\phi: \mathcal{O}_{k+1} \rightarrow \mathcal{K}(\mathcal{X}_\eta)$ by compact operators; the out-covering bijection on out-arcs is exactly what makes ϕ satisfy (CK2), where the sum-over-lifts homomorphism of [VIII] had failed. The pair $(\mathcal{X}_\eta, \phi)$ is an injective, full, proper correspondence, so it defines a canonical class $[\mathcal{X}_\eta] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$, and its K_0 -index map is precisely the head pushforward η_* ; as a right module $[\mathcal{X}_\eta] = q[1_{\mathcal{O}_k}]$. The correspondence is not invertible in KK : its index η_* has kernel the 2-torsion of $\text{Jac}(Y_{k+1})$, of order $q^{n_k(q-1)-1}$. The two families of morphisms organize the levels into a small KK -category with objects $\{\mathcal{O}_k\}$: alongside the upward homomorphisms $[\varphi_k]$ (K_0 -map τ^*) sit the downward correspondences $[\mathcal{X}_\eta]$ (K_0 -map η_*), and their Kasparov products are the Hecke operators — $[\varphi_k] \otimes [\mathcal{X}_\eta]$ acts on $K_0(\mathcal{O}_k)$ as A_k^t and $[\mathcal{X}_\eta] \otimes [\varphi_k]$ on $K_0(\mathcal{O}_{k+1})$ as A_{k+1}^t . This is the C^* -bivariant form of the degeneracy calculus $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ of [V], and it completes the two-sided operator-algebraic picture of the tower. Everything is verified exactly on the K_4 tower at levels $k \leq 3$.

STATUS OF THE RESULTS

Every numbered statement is proved. The C^* -algebraic inputs are the standard theory of C^* -correspondences and their Kasparov classes [2, 1], the graph-algebra K -theory and gauge-invariant uniqueness of [4], and Kaminker–Putnam Spanier–Whitehead duality [3] (cited once, for context, in Remark 3.3); the construction of $\mathcal{X}_\eta$ and the computation of its index are done in full. The in-covering homomorphism φ_k and the matrix identities of the tower are quoted from [VIII] and re-verified. Every finite identity was machine-verified exactly on the K_4 tower, $q = 2$, at the transitions $k + 1 \rightarrow k$ for $k \leq 3$ (shadow graphs on 24, 48, 96 vertices; Table 1). Section 6 records what is solved, what stays open, and where the specification is corrected.

1. SETTING: THE TWO DEGENERACY MAPS AND THE MIRROR PROBLEM

We keep the conventions of [VIII]. Y_0 is a finite connected $(q+1)$ -regular graph, $q \geq 2$; for $k \geq 1$, Y_k is the Iwahori path graph, a strongly connected q -regular digraph on $n_k = n(q+1)q^{k-1}$ vertices with $Y_{k+1} = L(Y_k)$ its line digraph, so $V_{Y_{k+1}} = \text{Arcs}(Y_k)$. The tail and head maps $\tau, \eta: Y_{k+1} \rightarrow Y_k$

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Paper XVI of the series; Papers I–XV are cited as [I]–[XV]. The verification script `head_correspondence.py` in the ancillary bundle reproduces every computational claim; its output is archived as `head_correspondence_output.txt`.

send an arc to its source and range; $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ are their pushforwards, $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ the pullbacks, with [V, Prop. 1.2]

$$(1.1) \quad \tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI, \quad \tau_* \Delta_{k+1} = \Delta_k \tau_*, \quad \eta_* \Delta_{k+1}^t = \Delta_k^t \eta_*,$$

$\Delta_k = qI - A_k$. The shadow graph $Y_k^\sharp$ has vertex set $V_{Y_k} \sqcup V_{Y_k}'$, the arcs of Y_k , one arc $v \rightarrow v'$, $c := q - 1$ return arcs $v' \rightarrow v$, and two loops at each v' ; its vertex matrix is $B_k = \begin{pmatrix} A_k & I \\ cI & 2I \end{pmatrix}$ and $\mathcal{O}_k := C^*(Y_k^\sharp)$ is a unital Kirchberg algebra with, in the convention $K_0(C^*(E)) = \text{coker}(1 - A_E^t)$ of [4, Thm. 7.16],

$$(1.2) \quad K_0(\mathcal{O}_k) = \text{coker}(I - B_k^t) \cong \mathbb{Z} \oplus \text{Jac}(Y_k), \quad K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z},$$

by the factorization $U(I - B_k^t)V = \Delta_k^t \oplus (-I)$ of [VIII, Thm. 1.4] with $U = \begin{pmatrix} I & -cI \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$.

[VIII] proved that τ extends to a graph morphism $\tau^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ that is an *in*-covering — bijective on the in-arcs at every vertex — and that in-coverings induce, by $p_v \mapsto \sum_{\tau^\sharp(w)=v} p_w$, $s_a \mapsto \sum_{\tau^\sharp(b)=a} s_b$, a canonical unital injective $*$ -homomorphism $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$ [VIII, Lem. 2.2, Thm. 2.5], whose K_0 -map is the pullback τ^* (up the tower) and whose K_1 -map is the identity.

The head map is the mirror. Its shadow extension $\eta^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ is an *out*-covering, bijective on out-arcs and q -to-one on in-arcs (Table 1, line (O)), and the sum-over-lifts formula *fails* to be a homomorphism: for two arcs $f \neq f'$ of $Y_{k+1}^\sharp$ over the same arc with the same range, the cross term $s_f s_{f'}^*$ does not vanish, so $(\sum_b s_b)$ is not a Cuntz–Krieger partial isometry [VIII, Rem. 2.4]. Yet on K_0 it is the head pushforward η_* , not τ_* , that descends:

$$(1.3) \quad \eta_* A_{k+1}^t = A_k^t \eta_*, \quad H_k := \begin{pmatrix} \eta_* & 0 \\ 0 & \eta_* \end{pmatrix} \text{ satisfies } H_k(I - B_{k+1}^t) = (I - B_k^t)H_k,$$

the first being the transpose of $\tau_* A_{k+1} = A_k \tau_*$ from (1.1); whereas $\text{diag}(\tau_*, \tau_*)$ does *not* intertwine $I - B^t$ [VIII, Thm. 2.5(iv)]. So the arrow the tower asks for — a morphism $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ realizing η_* downward — exists on K_0 but cannot be a homomorphism, since additionally $H_k \mathbf{1} = q\mathbf{1}$ makes $\eta_*[1_{\mathcal{O}_{k+1}}] = q[1_{\mathcal{O}_k}]$ (a unit class cannot map to q times a unit class). The resolution is that the morphism is a C^* -correspondence.

2. THE OUT-COVERING CORRESPONDENCE

We recall that a C^* -correspondence ${}_A \mathcal{X}_B$ is a right Hilbert B -module $\mathcal{X}$ with a $*$ -homomorphism $\phi: A \rightarrow \mathcal{L}(\mathcal{X})$; if $\mathcal{X}$ is finitely generated projective over a unital B then $\mathcal{L}(\mathcal{X}) = \mathcal{K}(\mathcal{X})$, and $(\mathcal{X}, \phi, 0)$ is an even Kasparov cycle defining $[\mathcal{X}] \in KK(A, B)$ whose product with $[e] \in K_0(A)$, $e = e^2 = e^* \in M_\infty(A)$, is $[\phi(e)\mathcal{X}] \in K_0(B)$ [2, 1].

Throughout this section $p := \eta^\sharp: F \rightarrow E$, $F = Y_{k+1}^\sharp$, $E = Y_k^\sharp$, an out-covering: $s^{-1}(w) \rightarrow s^{-1}(p(w))$ is a bijection for every $w \in F^0$. Write $\{p_v, s_e\}$ for the generators of $\mathcal{O}_k = C^*(E)$ and $\{q_w, t_f\}$ for those of $\mathcal{O}_{k+1} = C^*(F)$.

Definition 2.1 (The head module). Let

$$\mathcal{X}_\eta := \bigoplus_{w \in F^0} p_{p(w)} \mathcal{O}_k,$$

the right Hilbert $\mathcal{O}_k$ -module of finitely supported families $\xi = (\xi_w)_{w \in F^0}$ with $\xi_w \in p_{p(w)} \mathcal{O}_k$, right action $(\xi \cdot a)_w = \xi_w a$, and $\mathcal{O}_k$ -valued inner product $\langle \xi, \zeta \rangle = \sum_{w \in F^0} \xi_w^* \zeta_w$. Let $e_w \in \mathcal{X}_\eta$ be the generator with w -component $p_{p(w)}$ and all others 0.

Lemma 2.2 (The module is finitely generated projective and full). $\mathcal{X}_\eta$ is a right Hilbert $\mathcal{O}_k$ -module with $\langle e_w, e_{w'} \rangle = \delta_{ww'} p_{p(w)}$, finite orthogonal generating set $\{e_w\}_{w \in F^0}$, and reconstruction

$\xi = \sum_w e_w \langle e_w, \xi \rangle$; hence $\mathcal{X}_\eta$ is finitely generated projective, $\mathcal{L}(\mathcal{X}_\eta) = \mathcal{K}(\mathcal{X}_\eta)$, and

$$[\mathcal{X}_\eta] = \sum_{w \in F^0} [p_{p(w)}] = q [1_{\mathcal{O}_k}] \in K_0(\mathcal{O}_k),$$

because every fibre of $\eta^\sharp$ has exactly q elements. $\mathcal{X}_\eta$ is full: $\overline{\langle \mathcal{X}_\eta, \mathcal{X}_\eta \rangle} = \mathcal{O}_k$.

Proof. $\mathcal{X}_\eta$ is an orthogonal direct sum of the corners $p_{p(w)}\mathcal{O}_k$, each a Hilbert $\mathcal{O}_k$ -module with $\langle a, b \rangle = a^*b$; the displayed inner products and reconstruction are immediate, so $\{e_w\}$ is a finite frame and $\mathcal{X}_\eta \cong \bigoplus_w p_{p(w)}\mathcal{O}_k$ is finitely generated projective with $[\mathcal{X}_\eta] = \sum_w [p_{p(w)}]$. The fibre of $\eta^\sharp$ over a Y -vertex v is $\{\text{arcs } f \text{ of } Y_k \text{ with } r(f) = v\}$, of size the in-degree q ; over v' it is the q shadow lifts likewise, so each vertex of E is hit q times and $\sum_w [p_{p(w)}] = q \sum_v [p_v] = q[1]$. Fullness: $\langle e_w, e_w \rangle = p_{p(w)}$ and $\{p_v\}$ generates $\mathcal{O}_k$ as an ideal (it contains the unit). $\square$

Theorem 2.3 (The left action). *There is a unique $*$ -homomorphism $\phi: \mathcal{O}_{k+1} \rightarrow \mathcal{K}(\mathcal{X}_\eta)$ with*

$$\phi(q_w) = \Theta_{e_w, e_w} \quad (\text{projection onto the } w\text{-summand}), \quad \phi(t_f) e_w = \delta_{w, r(f)} e_{s(f)} \cdot s_{p(f)}.$$

It is injective. The verification of the Cuntz–Krieger relation (CK2) for the operators $\phi(t_f)$ uses precisely that $\eta^\sharp$ is an out-covering.

Proof. Define $V_f := \phi(t_f)$ by the stated formula: V_f maps the $r(f)$ -summand $p_{p(r(f))}\mathcal{O}_k$ into the $s(f)$ -summand by left multiplication by $s_{p(f)}$, and kills the other summands. Since p is a graph morphism, $p(f)$ runs $p(s(f)) \rightarrow p(r(f))$, so $s_{p(f)} = p_{p(s(f))} s_{p(f)} p_{p(r(f))}$ and V_f does land in $p_{p(s(f))}\mathcal{O}_k$; each V_f is adjointable with $V_f^* \xi = \delta$ -shift back by $s_{p(f)}^*$, hence $V_f \in \mathcal{K}(\mathcal{X}_\eta)$ (finite rank in the frame $\{e_w\}$). Now

$$V_f^* V_f \xi = s_{p(f)}^* s_{p(f)} \xi_{r(f)} = p_{r(p(f))} \xi_{r(f)} = p_{p(r(f))} \xi_{r(f)} = \xi_{r(f)} \quad (\xi \in \mathcal{X}_\eta),$$

using (CK1) in $\mathcal{O}_k$ and $r(p(f)) = p(r(f))$; so $V_f^* V_f = \Theta_{e_{r(f)}, e_{r(f)}} = \phi(q_{r(f)})$, which is (CK1) for ϕ . For (CK2), fix $w \in F^0$ and compute $\sum_{s(f)=w} V_f V_f^*$. Each such $V_f V_f^*$ acts on the $w = s(f)$ -summand by $\xi \mapsto s_{p(f)} s_{p(f)}^* \xi$, so

$$\sum_{s(f)=w} V_f V_f^* \Big|_{w\text{-summand}} = \sum_{s(f)=w} s_{p(f)} s_{p(f)}^* = \sum_{s(e)=p(w)} s_e s_e^* = p_{p(w)},$$

where the middle equality is the out-covering property: $f \mapsto p(f)$ is a *bijection* from $\{f : s(f) = w\}$ onto $\{e : s(e) = p(w)\}$, so the sum reindexes exactly, and the last equality is (CK2) in $\mathcal{O}_k$. The right side is the identity of the w -summand, i.e. $\phi(q_w)$; and $V_f V_f^*$ kills every other summand. Hence $\sum_{s(f)=w} \phi(t_f) \phi(t_f)^* = \phi(q_w)$, which is (CK2) for ϕ . The universal property of $C^*(F)$ gives the $*$ -homomorphism ϕ . Injectivity: $\phi(q_w) \neq 0$ for every w (its range $p_{p(w)}\mathcal{O}_k \neq 0$), and $\mathcal{O}_{k+1}$ is simple, so $\ker \phi = 0$.

The out-covering hypothesis enters at exactly one point — the reindexing $\{p(f) : s(f) = w\} = \{e : s(e) = p(w)\}$ in (CK2) — and nowhere else. For the in-covering $\tau^\sharp$ this reindexing holds on *in*-arcs, which is why $\tau^\sharp$ gave a homomorphism into $\mathcal{O}_{k+1}$ rather than a correspondence; for $\eta^\sharp$ it is the out-arcs that biject, and the natural home is $\mathcal{K}(\mathcal{X}_\eta)$. The cross terms $s_{p(f)} s_{p(f')}^*$, $s_{p(f')} s_{p(f)}^*$ that obstructed the sum-over-lifts homomorphism of [VIII] are here mutually orthogonal (distinct out-arcs $p(f) \neq p(f')$ of $p(w)$), so they add rather than interfere. $\square$

Definition 2.4 (The head correspondence and its Kasparov class). $\mathcal{E}_\eta := (\mathcal{X}_\eta, \phi)$ is an injective, full, proper C^* -correspondence from $\mathcal{O}_{k+1}$ to $\mathcal{O}_k$; its Kasparov class is

$$[\mathcal{E}_\eta] := [(\mathcal{X}_\eta, \phi, 0)] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k).$$

3. THE INDEX IS THE HEAD PUSHFORWARD

Theorem 3.1 (K_0 -index of the head correspondence). *The map $\otimes_{\mathcal{O}_{k+1}}[\mathcal{E}_\eta]: K_0(\mathcal{O}_{k+1}) \rightarrow K_0(\mathcal{O}_k)$ is the head pushforward: under (1.2) it is induced by $H_k = \text{diag}(\eta_*, \eta_*)$, i.e. the descent of $\eta_*: \text{coker } \Delta_{k+1}^t \rightarrow \text{coker } \Delta_k^t$ (1.3). On generators $[q_w] \mapsto [p_{\eta^\sharp(w)}]$; the K_1 -map is multiplication by q ; and $[\mathcal{E}_\eta] \otimes [1_{\mathcal{O}_k}]^\vee$ recovers $[\mathcal{X}_\eta] = q[1_{\mathcal{O}_k}]$.*

Proof. For a vertex projection $q_w \in \mathcal{O}_{k+1}$, $\phi(q_w)\mathcal{X}_\eta$ is the w -summand $p_{p(w)}\mathcal{O}_k$, of class $[p_{p(w)}] = [p_{\eta^\sharp(w)}]$ in $K_0(\mathcal{O}_k)$; since the classes $[q_w]$ generate $K_0(\mathcal{O}_{k+1})$ and the product is additive, the index map sends $[q_w] \mapsto [p_{\eta^\sharp(w)}]$, which on the generators $e_w \mapsto e_{\eta^\sharp(w)}$ of $\text{coker}(I - B^t)$ is the matrix $H_k = \text{diag}(\eta_*, \eta_*)$. That H_k descends to $\text{coker}(I - B_{k+1}^t) \rightarrow \text{coker}(I - B_k^t)$ is (1.3); conjugating by U of [VIII, Thm. 1.4] (which commutes with diag blocks) turns it into η_* on $\text{coker } \Delta^t$. The K_1 -map: $K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z}(\mathbf{1}, -\mathbf{1})$ and $H_k(\mathbf{1}, -\mathbf{1}) = q(\mathbf{1}, -\mathbf{1})$ (Table 1, line (K)). The last statement is Lemma 2.2. $\square$

Proposition 3.2 (The correspondence is not invertible). *$[\mathcal{E}_\eta]$ is not a KK -equivalence. Its index η_* on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ is surjective with kernel the p -torsion subgroup $\text{Jac}(Y_{k+1})[p]$ when $q = p$ is prime, of order $p^{n_k(p-1)-1}$; equivalently $\mathcal{X}_\eta$ is not a Morita equivalence bimodule ($\phi(\mathcal{O}_{k+1}) \subsetneq \mathcal{K}(\mathcal{X}_\eta)$).*

Proof. By reversal symmetry (Remark 3.3) η_* on $\text{Tor coker } \Delta_{k+1}^t \rightarrow \text{Tor coker } \Delta_k^t$ is the mirror of the tail norm τ_* on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ of [VIII, Thm. 2.8]: surjective with kernel the p -torsion of order $p^{n_k(p-1)-1}$ (Table 1, line (N)). A KK -equivalence induces isomorphisms on K_* ; η_* is not injective, so $[\mathcal{E}_\eta]$ is not invertible. Were $\mathcal{X}_\eta$ an imprimitivity bimodule, ϕ would be onto $\mathcal{K}(\mathcal{X}_\eta)$ and the index an isomorphism. $\square$

Remark 3.3 (Reversal symmetry). Let ρ_k be the path-reversal permutation of V_{Y_k} (reverse each non-backtracking path). Then $\rho_k A_k \rho_k^{-1} = A_k^t$ and $\rho_k \tau_* = \eta_* \rho_{k+1}$ (Table 1, line (R)): reversing all arrows exchanges $\tau \leftrightarrow \eta$, $\Delta \leftrightarrow \Delta^t$, $\mathcal{O}_B \leftrightarrow \mathcal{O}_{B^t}$, and carries the in-covering theory of [VIII] for (τ, Δ) to the out-covering theory of this paper for (η, Δ^t) . At the algebra level reversal is Kaminker–Putnam Spanier–Whitehead duality [3], $\mathcal{O}_B$ and $\mathcal{O}_{B^t}$ dual, which exchanges K_0 with K^1 ; we use only the elementary permutation identity, not the duality, and cite [3] for context.

4. KASPAROV PRODUCTS AND THE HECKE OPERATORS

Write $[\varphi_k] \in KK(\mathcal{O}_k, \mathcal{O}_{k+1})$ for the class of the tail homomorphism (K_0 -map τ^* , upward) and $[\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ for the head correspondence (K_0 -map η_* , downward).

Theorem 4.1 (Round trips are Hecke operators). *The Kasparov products*

$$[\varphi_k] \otimes_{\mathcal{O}_{k+1}} [\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_k, \mathcal{O}_k), \quad [\mathcal{E}_\eta^{(k)}] \otimes_{\mathcal{O}_k} [\varphi_k] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_{k+1})$$

induce on K_0 the maps

$$\eta_* \tau^* = A_k^t \text{ on } K_0(\mathcal{O}_k), \quad \tau^* \eta_* = A_{k+1}^t \text{ on } K_0(\mathcal{O}_{k+1}),$$

i.e. the level Hecke operators. These are the C^ -bivariant form of the degeneracy identities $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ of [V, Prop. 1.2].*

Proof. The K_0 -functor takes Kasparov products to composites, so $\otimes([\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]) = (\otimes[\mathcal{E}_\eta^{(k)}]) \circ (\otimes[\varphi_k]) = H_k \circ T^k$ with $T^k = \text{diag}(\tau^*, \tau^*) = K_0(\varphi_k)$ (Theorem 3.1, [VIII, Thm. 2.5]). Block-wise $H_k T^k = \text{diag}(\eta_* \tau^*, \eta_* \tau^*)$, and $\eta_* \tau^* = (\tau_* \eta^*)^t = A_k^t$ by transposing (1.1); similarly $T^k H_k = \text{diag}(\tau^* \eta_*, \tau^* \eta_*)$ with $\tau^* \eta_* = (\eta^* \tau_*)^t = A_{k+1}^t$ (Table 1, line (H)). Both descend to $\text{coker}(I - B^t)$, since each factor does. $\square$

Remark 4.2 (The opposite side and the full calculus). The other two degeneracy maps, τ_* and η^* , descend not on K_0 but on the Bowen–Franks group $\text{coker}(I - B) \cong \text{coker } \Delta$, the K_0 of the opposite algebra $\mathcal{O}_k^{\text{op}} \cong C^*(\overline{Y_k^\sharp})$: there $\text{diag}(\tau_*, \tau_*)$ and $\text{diag}(\eta^*, \eta^*)$ intertwine $I - B$, with composites A_k and A_{k+1} (Table 1, line (B)). Thus all four degeneracy maps of [V] are realized, two by KK -classes on the K_0 -side (τ^* up, η_* down) and two by their reversal-dual partners on the Bowen–Franks side; the “ qI ” identities $\tau_*\tau^* = \eta_*\eta^* = qI$ pair one map from each side and are the Pontryagin adjointness of [VIII, Lem. 2.1]. The correspondence $\mathcal{E}_\eta$ supplies the one arrow that was missing: the canonical downward morphism inducing η_* .

Definition 4.3 (The KK -category of the tower). Let $\mathcal{T}$ be the small category enriched in KK (a subcategory of the Kasparov category KK) with objects $\{\mathcal{O}_k : k \geq 1\}$ and morphisms generated under Kasparov product and $\mathbb{Z}$ -linear combination by $[\varphi_k] \in KK(\mathcal{O}_k, \mathcal{O}_{k+1})$ and $[\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ for all k .

Corollary 4.4 (Structure of $\mathcal{T}$). *In $\mathcal{T}$ the endomorphism $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]$ of $\mathcal{O}_k$ acts on $K_0(\mathcal{O}_k)$ as the Hecke operator A_k^t , and $[\mathcal{E}_\eta^{(k)}] \otimes [\varphi_k]$ on $\mathcal{O}_{k+1}$ as A_{k+1}^t ; the diagonal composite $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}] \otimes \dots$ over a τ -then- η round trip realizes, on K_0 , the polynomial algebra generated by the Hecke operators of the tower. No morphism of $\mathcal{T}$ is invertible except the identities: $[\varphi_k]$ has index τ^* (injective, not onto: cokernel the new p -torsion) and $[\mathcal{E}_\eta^{(k)}]$ has index η_* (onto, not injective), so $\mathcal{T}$ is a genuinely one-way-plus-one-way category, the operator-algebraic shadow of the two-sidedness of the pro-module X_∞ of [V].*

Proof. Theorems 3.1, 4.1; injectivity/surjectivity from [VIII, Thm. 2.5] and Proposition 3.2. $\square$

5. THE VERIFICATION BATTERY

check	transitions	result
(O) $\eta^\sharp$ morphism, bijective on out-arcs, q -to-one on in-arcs; $\tau^\sharp$ the mirror in-covering	$k \leq 3$	exact
(R) $\rho_k A_k \rho_k^{-1} = A_k^t$, $\rho_k \tau_* = \eta_* \rho_{k+1}$	$k \leq 3$	exact
(K) $H_k(I - B_{k+1}^t) = (I - B_k^t)H_k$; $\text{diag}(\tau_*, \tau_*)$ does not; $H_k \mathbf{1} = q\mathbf{1}$; $H_k(\mathbf{1}, -\mathbf{1}) = q(\mathbf{1}, -\mathbf{1})$	$k \leq 3$	exact
(H) $H_k T^k = \text{diag}(A_k^t)$, $T^k H_k = \text{diag}(A_{k+1}^t)$; witness $\text{diag}(A^t - q) = (I - B^t)VN$	$k \leq 3$	exact
(B) opposite side: $\text{diag}(\tau_*, \tau_*)$, $\text{diag}(\eta^*, \eta^*)$ descend on $\text{coker}(I - B)$; composites A_k, A_{k+1}	$k \leq 3$	exact
(N) η_* on $\text{Tor } \text{coker } \Delta^t$: surjective, $ \ker = 2^{11}, 2^{23}, 2^{47} = 2^{n_k(q-1)-1}$; $\text{rank}_{\mathbb{F}_2} A_{k+1} = n_k$	$k \leq 3$	exact
(X) $\text{Ext}(K_0(\mathcal{O}_{k+1}), K_1(\mathcal{O}_k)) = \text{Jac}(Y_{k+1})$ (lift ambiguity); $\text{Ext}(K_0(\mathcal{O}_k), K_1(\mathcal{O}_k)) = \text{Jac}(Y_k)$	$k \leq 3$	exact

TABLE 1. Exact integer verification (`head_correspondence.py`) on the K_4 tower, $q = 2$, at the transitions $k + 1 \rightarrow k$, $k \leq 3$; shadow matrices of size 24, 48, 96, 192. Jac data as in [VIII, Table 1].

6. SCOPE, RESIDUE, CORRECTIONS

Remark 6.1 (Solved, open, impossible). *Solved:* the open item [VIII, Rem. 5.1(a)] — a canonical C^* -correspondence $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ induced by the out-covering $\eta^\sharp$ with K_0 -map η_* . It is $\mathcal{E}_\eta = (\mathcal{X}_\eta, \phi)$ of Definition 2.4: a finitely generated projective, full, injective proper correspondence whose class in $KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ induces η_* (Theorem 3.1), completing the two-sided realization of the degeneracy calculus (Theorem 4.1, Remark 4.2) and organizing the levels into the KK -category $\mathcal{T}$ (Definition 4.3, Corollary 4.4). *Open:* (a) the Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{X}_\eta}$ of the head correspondence, and whether the inductive system it generates gives a second model of the limit $\mathcal{O}_\infty$ of [VIII] dual to the first; (b) an explicit unbounded (spectral-triple) representative of $[\mathcal{E}_\eta]$ pairing with the

$\mathcal{L}$ -invariant KK -classes of [XIV]; (c) the Hecke-module structure of $K_0(\mathcal{O}_\infty)$ under the operators $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]$ of Corollary 4.4, and with it [V, Prob. 5.1]. *Impossible*: a unital $*$ -homomorphism $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ inducing η_* (Section 1: $\eta_*[1] = q[1]$); and, as throughout the series, a KK -equivalence between adjacent levels (Proposition 3.2).

Remark 6.2 (Corrections to the task specification). (1) The correspondence’s right module is not “freely generated by the graph’s generators”; it is the finitely generated projective module $\mathcal{X}_\eta = \bigoplus_w p_{\eta^\#(w)} \mathcal{O}_k$, a sum of corners, of K -theory class $q[1]$. (2) Positive-definiteness and self-adjointness of the inner product are automatic (the inner product is $\sum_w \xi_w^* \zeta_w$, a sum of positives) and use no Cuntz–Krieger relation; what (CK2) — together with the out-covering bijection on *out*-arcs — establishes is that the *left action* ϕ is a $*$ -homomorphism (Theorem 2.3). This is the one place the geometry is used, and it is on out-arcs, not on the (CK1) side as the specification suggested. (3) The left action lands in $\mathcal{K}(\mathcal{X}_\eta)$, not in $\mathcal{O}_k$; that is exactly why the object is a correspondence and not a homomorphism, and why it exists where the sum-over-lifts homomorphism of [VIII] could not. (4) The head pushforward η_* , not τ_* , is the map the downward correspondence induces; τ_* descends on the opposite (Bowen–Franks) side, not on K_0 (Remark 4.2). (5) “All six degeneracy identities in KK ” is made precise as the Kasparov products of Theorem 4.1 and the opposite-side identities of Remark 4.2: the two Hecke operators are the two round-trip products, and $\tau_* \tau^* = \eta_* \eta^* = qI$ are cross-side adjointness identities, not products of two K_0 -side classes.

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Unrefereed preprint. Every numbered statement is proved; the correspondence $\mathcal{X}_\eta$, its left action, and its index are constructed and computed in full, and the tower matrix identities are quoted from [VIII] and re-verified exactly on K_4 . The bivariant inputs — C^* -correspondences define KK -classes, and K_0 turns Kasparov products into composites — are standard [2, 1]; [3] is cited for context only. Bibliographic data of [2, 1, 4, 3] were checked against publisher or indexing pages.

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